Computer Graphics

Viewing Transforms

Dr. Hardik Jain
Assistant Professor
Department of Computer Science and Engineering
IIT Jodhpur

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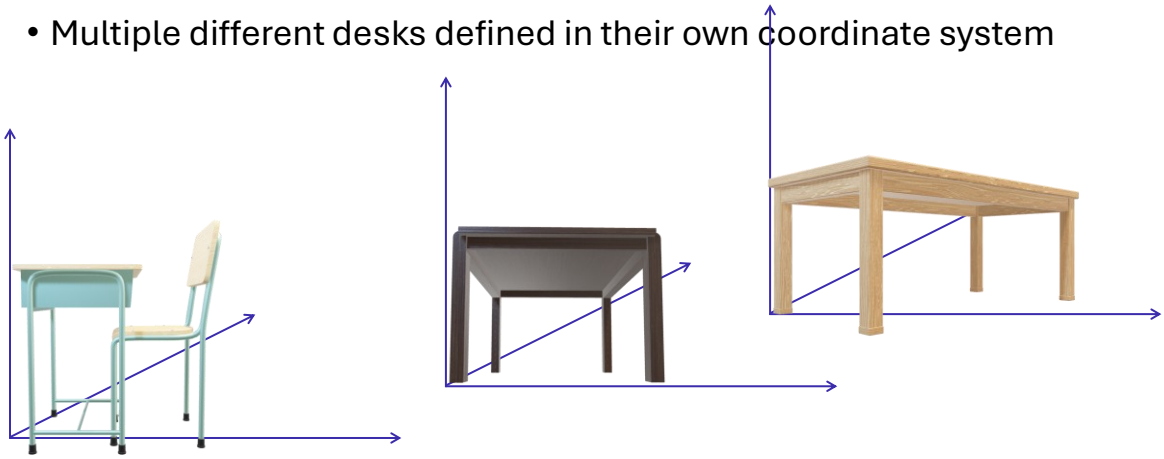
Aligning Multiple Coordinate Systems

- A desk defined in its own coordinate system
- Multiple different desks defined in their own coordinate system



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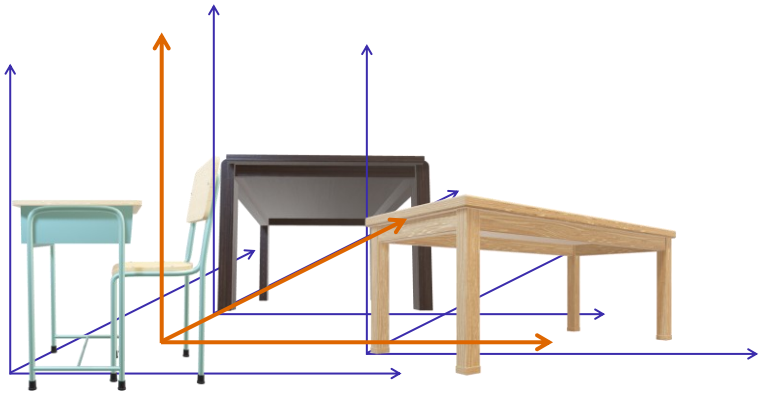
Aligning Multiple Coordinate Systems

- Place them all in a common World Coordinate (WC) system

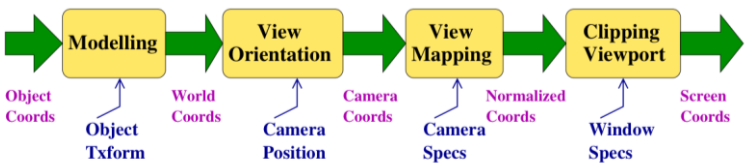


Different Coordinates

- **Object Reference**: Object is described in an internal coordinate frame called ORC.
- **World**: Common reference frame to describe different objects, called WC.
- **Camera/View Reference**: Describe with respect to the current camera position/orientation, called VRC.
- Ultimately, how the scene appears to the camera determines the image produced
- Goal of Computer Graphics: describe the scene in the camera coordinate frame

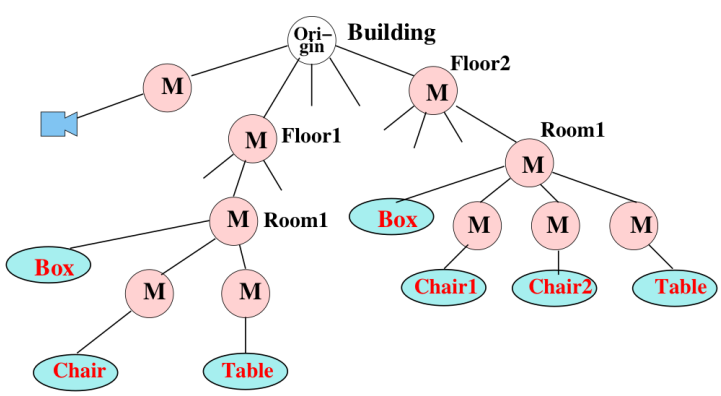
3D Graphics Pipeline

- Objects are specified in their own coordinate system and placed in the world coordinate frame.
- Camera is also placed in the world coordinate frame.
- Camera-to-world geometry is first projected to normalized coordinates and then to screen.



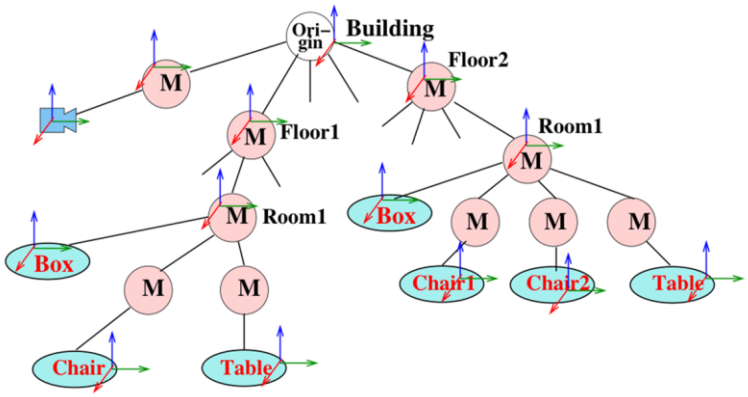
A Building Model

Hierarchical Model with root representing whole scene



A Building Model

Each Matrix M aligns parent frame to child frame



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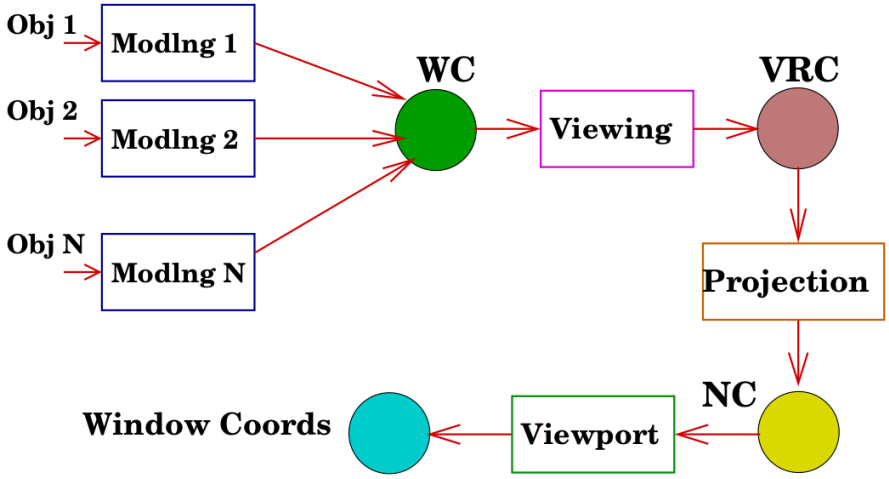
3D Graphics Pipeline



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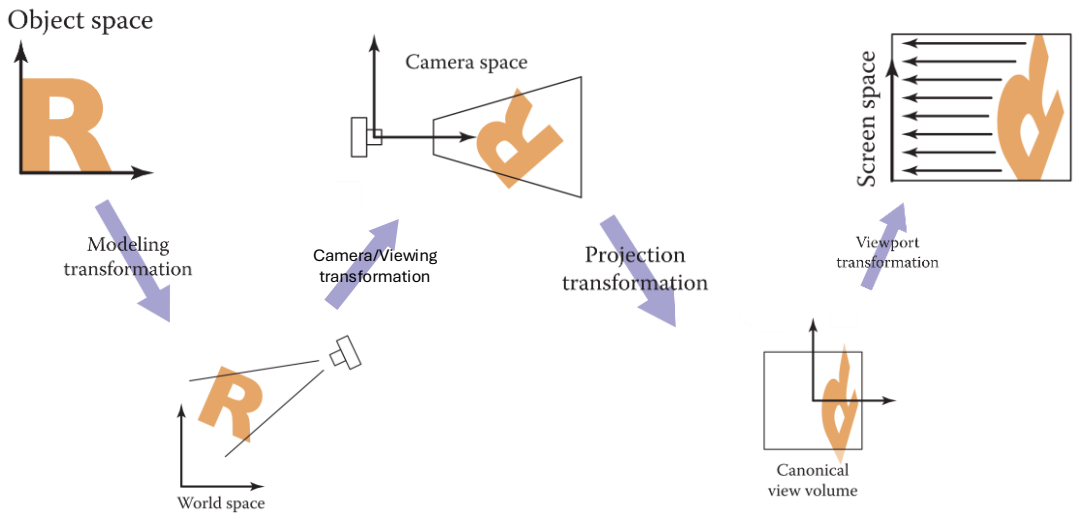
Different Coordinate Systems

- **Object Reference**: Object is described in an internal coordinate frame called ORC.
- **World**: Common reference frame to describe different objects, called WC.
- **Camera/View Reference**: Describe with respect to the current camera position/orientation, called VRC.
- **Normalized Projection**: A standard space from which projection is easy, called NPC.
- **Screen**: Coordinates in the output device space.

Transformations

- **Modelling**: Convert from object coordinates to world coordinates (ORC to WC).
- **View Orientation or Viewing**: From world coordinates to camera coordinates (WC to VRC).
- **View Mapping or Projection**: From VRC to Normalized Coordinates (NC).
- **Viewport**: From NC to window coordinates.

Transformation Visualization



Fundamental of Computer Graphics , 4e, Steve Marschner & Peter Shirley, Page 141.

Modelling and Viewing Transforms

- Transform points from object coordinates (ORC) to world coordinates (WC) to camera coordinates (VRC)
- A series of transformations for each object or point

$$P_{VRC} = | V | M | P_{ORC}$$

VRC WC ORC

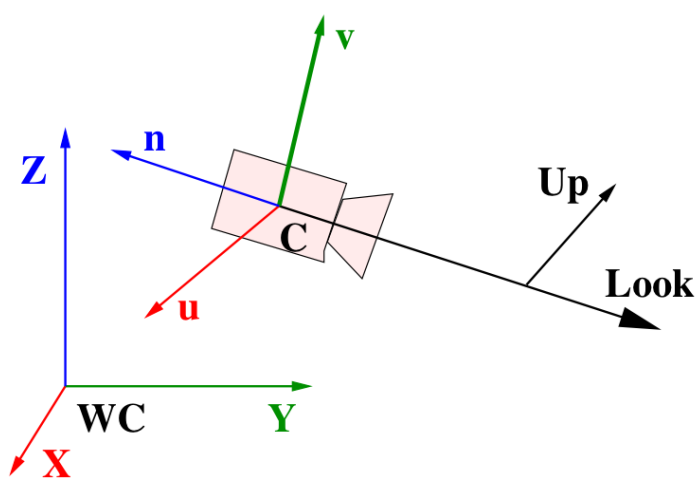
- **Goal**: Evaluate the coordinates of each point/line/triangle with respect to the camera

View Orientation

- Placing the camera in the world and orienting it right.
- Has 6 degrees of freedom: 3 for position and 3 for orientation.
- **Goal**: Transform points expressed in WC to VRC.
- Let u, v, n be the VRC or camera coordinate axes
- Viewing Transformation can be specified in many ways.
- Commonly using: Camera location, Look point, and Up direction.

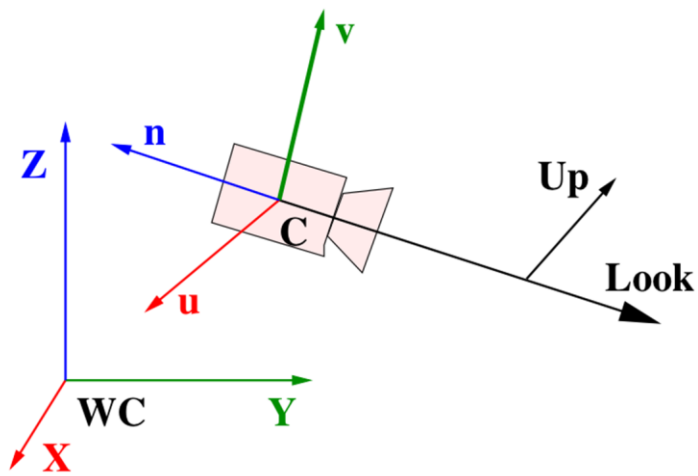
Placing Camera in the World

- Camera-center, Look-point and Up-vector specified in the world coordinates
- Object description in the WC needs to be transferred to VRC
- This conversion of object descriptions is equivalent to a sequence of transformations that superimposes the VRC onto the WC



Placing Camera in the World

- Translate the viewing-coordinate origin to the origin of the world coordinate system.
- Apply rotations to align the x_{view}, y_{view}, and z_{view} axes with the world x_w, y_w and z_w axes, respectively.



Placing Camera in the World

Translation Transformation

- The viewing-coordinate origin is at world position $\mathbf{P_0} = (x_0, y_0, z_0)$. Therefore, the matrix for translating the viewing origin to the world origin is

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -x_0 \\ 0 & 1 & 0 & -y_0 \\ 0 & 0 & 1 & -z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation Transformation

- The unit vectors $\mathbf{u}, \mathbf{v}, and\ \mathbf{n}$ can form the composite rotation matrix that superimposes the viewing axes onto the world frame.

$$\mathbf{R} = \begin{bmatrix} u_x & u_y & u_z & 0 \\ v_x & v_y & v_z & 0 \\ n_x & n_y & n_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

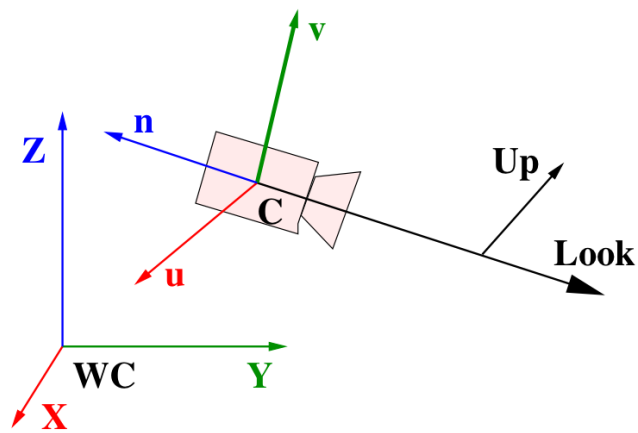
Transformation from World to Viewing Coordinates. Page309, Computer Graphics with OpenGL, 4e by Hearn, Baker & Carithers

Placing Camera in the World

Z axis

- Rotate to align Z-axis to —(Look Vector) or —L
- $\hat{l} = \frac{\bar{L}}{|\bar{L}|}, \hat{t} = \frac{\bar{U}}{|\bar{U}|}$
- Third row of the rotation matrix:

$$n = -\hat{l}$$

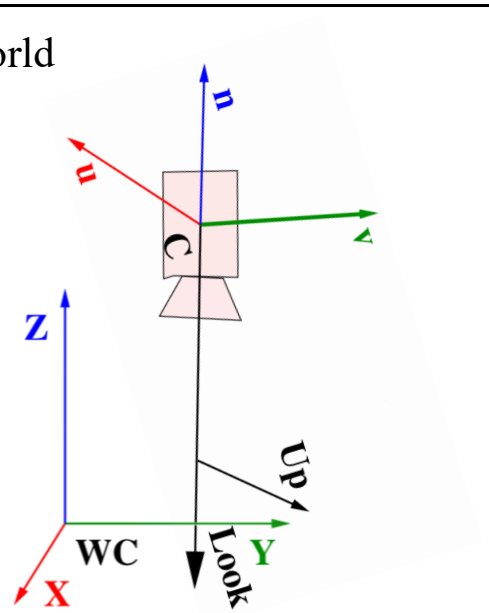


Placing Camera in the World

X & then Y axis

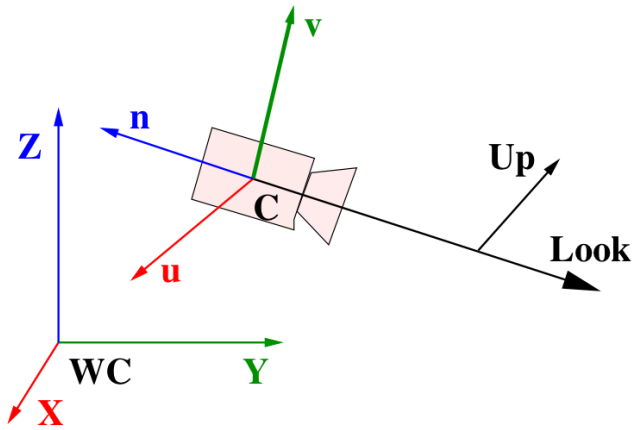
- Up vector needn't be orthogonal to the look vector.
- The $\bar{L}$ and $\bar{U}$ vectors define the "vertical" plane. A plane in the world that projects to a vertical line in the image. Or the camera's vn plane.

$$u = \frac{\hat{t} \times n}{|\hat{t} \times n|}$$
$$v = n \times u$$



Viewing Transform

- $V = R \cdot T$
- $P_{VRC} = V\, P_{WC}$



Modelling and Viewing

- Modelling and Viewing are not truly independent.
- What ultimately matters is only the relative geometry between the camera and the object(s).
- What we want is the description of each point in VRC, with respect to the camera.
- It is convenient to think of each object being placed in a WC and then the WC being transformed to VRC.
- Thus, each object has its modelling matrix. The scene has one viewing matrix.

Thank You

Anonymous Course Feedback